

**TABLE 8.6    What Is the Probability That Spending Is Sustainable?  
(Age 75; Random Lifespan and Random Returns Under a  
Fixed Spending Rate)**

| Allocation to<br>Equity | Needs-To-Wealth Ratio |       |       |       |       |
|-------------------------|-----------------------|-------|-------|-------|-------|
|                         | 5.0%                  | 6.0%  | 7.0%  | 8.0%  | 9.0%  |
| 0%                      | 86.4%                 | 79.6% | 72.2% | 64.6% | 57.0% |
| 20%                     | 90.0%                 | 84.4% | 78.0% | 71.0% | 63.9% |
| 40%                     | 91.7%                 | 86.9% | 81.2% | 75.0% | 68.4% |
| 60%                     | 92.2%                 | 87.8% | 82.6% | 76.9% | 70.8% |
| 80%                     | 91.8%                 | 87.5% | 82.6% | 77.2% | 71.5% |
| 100%                    | 90.7%                 | 86.4% | 81.5% | 76.4% | 71.0% |

Simulation assumptions: Median remaining lifetime = 10.7 yrs;  
Equity: expected returns =7%, volatility = 20%; Cash: return =1.3%  
Source: Moshe Milevsky and the IFID Centre, 2008.

As the 65-year-old moves into a more aggressive allocation to equity in Table 8.4, the probability of sustainability, surprisingly, increases. For example, with a 60% allocation to equities and a needs-to-wealth ratio of 8%, the retiree has a sustainability probability of 55.8%. While these odds of success are still fairly low, the higher average long-term rate of return counteracts the effects of the equity investment’s short-term volatility, yielding higher “sustainability odds.”

A few things should be immediately evident from the three tables. First, retiring at older ages leads to sustainability probabilities that are uniformly higher at all asset allocation levels. For example, under a 100% equity allocation at age 75, the sustainability probability numbers range from 71% to 90.7%, much higher (and much better) than the 51.7% to 81.1% range we see for a 65-year-old in the same situation (in Table 8.4) or the 39.8% to 73.3% for the 55 year-old (in Table 8.5). The reason is quite simple: You are 10 years older and, therefore, have 10 fewer years of consumption ahead of you. The odds of outliving your money, therefore, must be lower.